

Crimson King Maple

Acer platanoides 'Crimson King'

Species background: A cultivated variety of Norway maple first discovered as a chance red-leaved seedling around 1937 at a nursery in Limburg, Belgium, then introduced to the United States in the late 1940s. Unlike most trees, its purple color isn't a fall event -- the leaves are purple from the moment they open in spring and stay that way all summer, making it easy to spot from blocks away.

This packet contains one full lesson for each grade band: K-2, 3-5, 6-8, and 9-12. Each includes a learning objective, standards connection, materials, teacher background, a step-by-step procedure, discussion questions, an assessment, and an extension activity.

GRADE BAND K-2

Crimson King Maple: Find Crimson King's purple leaves and describe the color.

Standards connection: NGSS K-LS1-1; National Core Arts Standards VA:Cr1 (K-2)

Materials: Coloring sheet, purple crayons.

Teacher background

Find a purple hand-shaped leaf instead of a green one. This tree doesn't wait for fall to turn purple -- it stays that color all summer!

Procedure

- 1 Engage: Ask students what color they expect leaves to be.
- 2 Explore: Look at this tree's purple leaves and compare to a green-leaved tree nearby.
- 3 Explain: Explain that grown-ups picked this tree out specifically for its unusual purple color.
- 4 Art: Find a purple leaf and color a small tree using only purple.
- 5 Evaluate: Student can describe this tree's leaves as purple, not green.

Discussion questions

What other purple things can you find in nature?

Assessment

Correctly identifies the leaf color as purple.

Extension

Compare this tree's purple to any other purple-leaved tree nearby.

Crimson King Maple: Explain that Crimson King was specifically selected by people for its color.

Standards connection: NGSS 3-LS3-1; National Core Arts Standards VA:Cr2 (3-5)

Materials: Coloring sheet, paint mixing (red, blue, white).

Teacher background

This tree is a Norway maple bred to stay purple all summer instead of just in fall. It was first discovered as a chance purple seedling at a nursery long ago.

Procedure

- 1 Engage: Ask what it means for a plant to be 'discovered' as unusual and then intentionally grown.
- 2 Explore: Mix red, blue, and white paint to try to match the tree's purple shade.
- 3 Explain: Explain that this happened by chance once, and then people chose to grow more trees just like it.
- 4 Art: This tree stays purple all year -- paint a small tree using only purple, testing how many different shades you can mix.
- 5 Evaluate: Student can explain in one sentence that this tree was chosen by people for its unusual color.

Discussion questions

Can you think of other plants or animals that people have specifically bred for an unusual trait?

Assessment

Correct one-sentence explanation of intentional selection.

Extension

Research one other plant variety that was discovered and then intentionally propagated.

Crimson King Maple: Explain cultivar selection as an example of human-directed plant breeding.

Standards connection: NGSS MS-LS3-2; National Core Arts Standards VA:Cr2/Re7 (6-8)

Materials: Coloring sheet, mixed paints for shade-matching.

Teacher background

Crimson King keeps high anthocyanin levels all season by human selection, not just at leaf-drop -- a real example of selective breeding for an ornamental trait. It was one of the earliest patented ornamental street trees in the US.

Procedure

- 1 Engage: What's the difference between a trait that appears naturally in a population versus one that people deliberately select and propagate?
- 2 Explore: Research what it means for a plant variety to be 'patented.'
- 3 Explain: Explain cultivar selection and how it differs from natural variation across a wild population.
- 4 Art: Create a monochromatic purple painting of the tree, mixing at least 5 distinct shades from red, blue, and white -- label which mix ratio made each shade.
- 5 Evaluate: Write a paragraph explaining cultivar selection using Crimson King as the example.

Discussion questions

Should there be limits on how much of one cultivar a city plants, given they're often genetically identical clones?

Assessment

Paragraph correctly explains cultivar selection with accurate use of the term.

Extension

Research another patented ornamental tree cultivar and its origin story.

Crimson King Maple: Evaluate the genetics of clonal propagation and why seed-grown offspring wouldn't stay purple.

Standards connection: NGSS HS-LS3-2; National Core Arts Standards Proficient-level Creating/Responding (9-12)

Materials: Coloring sheet, research sources, color-mixing materials.

Teacher background

Crimson King is a clonally propagated cultivar selected for consistently high anthocyanin expression; research how cultivars are propagated by grafting rather than seed to preserve a specific genetic trait, and explain why seed-grown offspring of this tree would likely not stay purple.

Procedure

- 1 Engage: If you planted a seed from a Crimson King maple, would the resulting tree also be purple? Why or why not?
- 2 Explore: Research how grafting propagates an exact genetic clone, unlike growing from seed.
- 3 Explain: Connect genetic recombination in seed-grown offspring to why the purple trait wouldn't reliably pass on.
- 4 Art: Research color theory around anthocyanin-rich cultivars in ornamental horticulture, then create a color-mixing chart documenting the exact ratios used to replicate this leaf's specific shade of purple.
- 5 Evaluate: Submit a written explanation of grafting vs. seed propagation and the color-mixing chart.

Discussion questions

What are the risks of a city relying heavily on grafted clones of a single cultivar, genetically speaking?

Assessment

Rubric covering genetic reasoning accuracy and precision of the color-mixing chart.

Extension

Compare Crimson King's clonal-propagation risk to ginkgo's all-male clonal street-tree population.

SOURCES & METHODOLOGY

Background research drawn from TreeWalk NYC's own species research. Lesson structure (objective, background, materials, procedure, assessment, extension) follows the format used by real, freely published environmental-education lessons, including the National Park Service's "Looking at Leaves" 5E lesson plan (Acadia Outdoor Classroom Collaborative), Penn State Extension's "Sorting and Classifying Leaves," and Project Learning Tree's leaf-activity guidance -- adapted and written fresh for this species, not copied from those sources. This is an educational pilot; standards references indicate topic alignment, not formal certification.